

COURSE STRUCTURE

Course Code	BCA30040				
Course Category	Program Major				
Course Title	Introduction to Cloud Computing				
Teaching Scheme	Lectures	Tutorial	Laboratory / Practical	Project	Total
Weekly load hours	2	-	-	-	2
Credits	2	-	-	-	2
Assessment Scheme Code	TT1				

Pr-requisites:

1. Basic knowledge of computers

Course Objectives:

1. To understand the concept of Cloud Computing.
2. To be familiar with components required for different types of Cloud Services.
3. To be familiar with different Cloud Deployment Model.

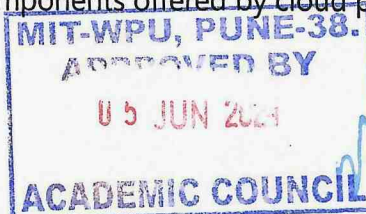
Course Outcomes:

Student will able to

1. Students will gain a comprehensive understanding of what cloud computing is, including its definition, characteristics, and essential components.
2. Students will learn about different cloud service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), including their characteristics, advantages, and use cases.
3. Students will be introduced to various cloud deployment models, including public cloud, private cloud, hybrid cloud, and community cloud, and understand when each deployment model is appropriate.
4. Students will develop skills in designing cloud architectures, including understanding the principles of scalability, elasticity, reliability, and security in cloud-based systems.
5. Students will become familiar with major cloud service providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), including their offerings, pricing models, and basic usage.
6. Students will gain practical experience by working with various cloud services such as virtual machines, storage solutions, databases, and networking components offered by cloud providers.

Course Contents:

Unit 1 : Understanding Cloud Computing: 6
Introduction to Cloud Computing



Characteristics of Cloud Computing
Evolution of Cloud Computing
Cloud Computing Use Cases
Benefits and Challenges of Cloud Computing

Unit 2 : Cloud Service Models: 6

Infrastructure as a Service (IaaS)
Platform as a Service (PaaS)
Software as a Service (SaaS)
Function as a Service (FaaS)
Comparison of Cloud Service Models

Unit 3 : Cloud Deployment Models: 6

Public Cloud
Private Cloud
Hybrid Cloud
Community Cloud
Choosing the Right Deployment Model

Unit 4 : Major Cloud Providers: 6

Amazon Web Services (AWS)
Microsoft Azure
Google Cloud Platform (GCP)
IBM Cloud
Oracle Cloud

Unit 5: Cloud Storage and Networking: 6

Cloud Storage Services
Content Delivery Networks (CDNs)
Virtual Private Clouds (VPCs)
Cloud Networking Concepts
Advanced Networking in the Cloud

Learning Resources:

Text Books/Reference Books:

- "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Ricardo Puttini, and Zaigham Mahmood
- "Cloud Computing For Dummies" by Judith Hurwitz, Robin Bloor, Marcia Kaufman, and Fern Halper
- "Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)" by Michael J. Kavis

Supplementary Reading:

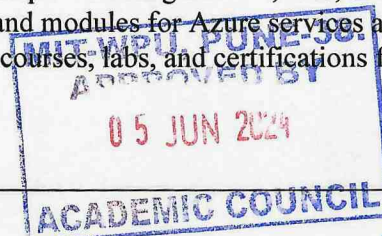
- AWS Documentation: Comprehensive documentation

Web Resources :

Weblinks:

- AWS Training and Certification: Offers free and paid training courses, labs, and certification exams for AWS.
- Microsoft Learn: Provides free learning paths and modules for Azure services and concepts.
- Google Cloud Training: Offers online training courses, labs, and certifications for Google Cloud Platform (GCP).

MOOCs: Online courses for self-learning
Courses by NPTEL and MIT Open



Courseware etc Pedagogy:

- Participative Learning,
- discussions,
- problem solving,
- assignments,
- Lab Assignment

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